Whats Cookin

Updated Development Plan

CS160A Course Project

Version Control Workflow

The project used a feature-based Git branching workflow, with each major area of functionality developed independently before being brought together for integration.

Development Process

1. Kumail developed the recipe-feature branch, covering recipe discovery, search, filtering, recipe details, and ingredient substitution against a temporary pantry data source.
2. Alexa developed the signup branch, covering Firebase Authentication, protected routing, and Firestore-backed pantry management.
3. After both major features were completed, Kumail created an integration branch to bring the two feature sets together.
4. The recipe system and the authentication/pantry systems were merged together on the integration branch, replacing the recipe feature's temporary pantry source with the real, live Firestore pantry.
5. Final debugging, UI consistency improvements, functional and stress testing, and demo-account preparation were performed on the integration branch.
6. The final stable version was merged into main for deployment.

Team Contributions

Kumail's Contributions

Recipe-feature Branch

- Created the recipe application foundation (project structure, routing, shared layout)
- Implemented the recipe homepage
- Integrated the Spoonacular API for live recipe data
- Created the recipe service layer for fetching and normalizing recipe data
- Implemented recipe cards
- Implemented recipe search
- Implemented recipe filtering (cuisine and time of day)
- Created the recipe details page
- Added missing-ingredient calculations
- Added ingredient substitution functionality
- Added API caching optimization to reduce Spoonacular API quota usage

Integration Phase

- Merged the recipe and authentication systems
- Connected pantry information with recipe features
- Debugged integration issues between the two feature branches
- Improved UI consistency across the merged application
- Prepared the demo account and testing environment
- Performed final testing
- Prepared the deployment-ready version

Alexa's Contributions

Signup Branch

- Implemented Firebase Authentication
- Created signup functionality
- Created login functionality
- Added authentication state management
- Implemented protected routes
- Connected Firebase Firestore
- Implemented pantry data storage
- Developed pantry functionality

Integration Phase

- Supported Firebase/pantry integration
- Ensured authentication and pantry workflows worked correctly with the recipe system
- Collaborated on the integration branch merging to resolve API and Firebase DB errors

Development Timeline

Phase	Major Tasks	Person Responsible	Status
Phase 1	Project Setup and Foundation, repository structure, routing, shared layout, and placeholder services established	Kumail, Alexa	Complete
Phase 2	Recipe Application Development, Spoonacular integration, recipe discovery, search, filtering, recipe details, missing-ingredient logic, and substitution feature built on the recipe-feature branch	Kumail	Complete
Phase 3	Authentication and Pantry Development, Firebase Authentication, protected routes, Firestore-backed pantry storage, and pantry management UI built on the signup branch	Alexa	Complete
Phase 4	Integration and Testing, recipe-feature and authentication/pantry branches merged on a dedicated integration branch, cross-feature debugging, UI consistency pass, functional and stress testing	Kumail, Alexa	Complete
Phase 5	Final Deployment and Documentation, demo account preparation, deployment-ready build, and project documentation	Kumail, Alexa	Complete

Completed Features

- Account creation, login, logout, and protected routing
- Personal pantry with quantity, unit, and optional expiry date per item
- Automatic expiry status tracking (Fresh / Expiring Soon / Expired)
- Live recipe discovery via the Spoonacular API (Recommended, Random, and Use Soon sections)
- Recipe search across titles and ingredients
- Recipe filtering by cuisine and time of day, combinable with search
- Recipe details page with servings, cuisine, ingredients, and instructions
- Ingredient availability display (available vs. missing) based on the live pantry
- Ingredient substitution feature limited to substitutes the user's pantry actually contains
- API response caching and request deduplication to reduce external API usage

Known Limitations

- Ingredient matching is name-based rather than quantity-aware, a pantry item is treated as available regardless of how much of it is on hand.
- The substitution database covers a fixed set of common ingredients rather than a comprehensive culinary substitution reference.
- Recipe data quality (images, instructions, servings) depends on what the Spoonacular API returns for a given recipe, which is not always complete.
- The application depends on an active internet connection for both authentication and recipe data; there is no offline mode.

Future Improvements

- Quantity-aware pantry matching, so a recipe is only marked cookable if enough of each ingredient is actually available
- Automatic pantry quantity deduction after a recipe is marked as cooked
- An expanded, more comprehensive ingredient substitution database
- Saved or favorited recipes for quick access
- Shopping-list generation from a recipe's missing ingredients